

User Management App..!

1) Update npm to Latest version

```
npm i npm@latest -g
```

2) Check if old version of Angular is already installed on your computer, if exist then remove it

```
npm uninstall -g @angular/cli  
npm cache clean  
npm install -g @angular/cli@7.3.9
```

Note: To install a specific version, you can use `npm install -g @angular/cli`

3) Create new project

```
ng new user-management-app
```

or
skip to install node modules while creating project

```
ng new user-management-app --skip-install  
npm install [run this command separately to load all dependencies]
```

4) Run your application to get started

```
cd user-management -app  
ng serve
```

5) Install bootstrap

```
npm install ngx-bootstrap bootstrap jquery
```

6) > Add styles and scripts in angular.json

In `angular.json`

```
"styles": [  
    "./node_modules/bootstrap/dist/css/bootstrap.min.css",  
    "src/styles.css"  
],  
"scripts": [  
    "node_modules/jquery/dist/jquery.min.js",  
    "node_modules/bootstrap/dist/js/bootstrap.min.js"  
],
```

```
"assets": [  
    "src/favicon.ico",  
    "src/assets",  
    "src/assets/images"  
]
```

7) Create Components

```
ng g component home  
ng g component login  
ng g component add-user  
ng g component edit-user  
ng g component list-user
```

Angular CLI Useful Commands

```
ng g component my-new-component  
ng g directive my-new-directive  
ng g pipe my-new-pipe  
ng g service my-new-service  
ng g class my-new-class  
ng g guard my-new-guard  
ng g interface my-new-interface  
ng g enum my-new-enum  
ng g module my-module
```

8) Configure Routing inside app.routing.ts

```
import { RouterModule, Routes } from '@angular/router';  
import { LoginComponent } from './login/login.component';  
import { AddUserComponent } from './add-user/add-user.component';  
import { ListUserComponent } from './list-user/list-user.component';  
import { EditUserComponent } from './edit-user/edit-user.component';
```

```

const routes: Routes = [
  { path: 'login', component: LoginComponent },
  { path: 'add-user', component: AddUserComponent },
  { path: 'list-user', component: ListUserComponent },
  { path: 'edit-user', component: EditUserComponent },
  {path : '', component : LoginComponent}
];

export const routing = RouterModule.forRoot(routes);

```

// OR

Inside app-routing.module.ts file

```

import { NgModule } from '@angular/core';
// Import RouterModule
import { Routes, RouterModule } from '@angular/router';
import { LoginComponent } from './login/login.component';
import { AddUserComponent } from './add-user/add-user.component';
import { ListUserComponent } from './list-user/list-user.component';
import { EditUserComponent } from './edit-user/edit-user.component';
import { HomeComponent } from './home/home.component';

```

// Each route maps a URL path to a component

```
const routes: Routes = [
```

```

  // This route specifies the route to redirect to if the path
  // is empty. In our case we are redirecting to /home
  // pathMatch property value can be full or prefix. For now we
  // will set it to full as we want to do a full match

```

```

  // { path: '',redirectTo:'/home',pathMatch:'full' },
  { path: '',component: HomeComponent },
  { path: 'home',component:HomeComponent},
  { path: 'login', component: LoginComponent },
  { path: 'add-user', component: AddUserComponent },
  { path: 'list-user', component: ListUserComponent },
  { path: 'edit-user', component: EditUserComponent },
  // {path : '', component : LoginComponent}, // or

```

```

  { path: '**', component: HomeComponent }
];

```

```
@NgModule({
  imports: [RouterModule.forRoot(routes)],
  exports: [RouterModule]
})
export class AppRoutingModule { }
```

9) Create a JSON server that serves the data.

```
npm install -g json-server
```

10) Run below command on terminal
json-server --watch db.json

11) Add below code inside db.json file

```
"users": [
  {
    "id": 1,
    "firstName": "Smita",
    "lastName": "Kumar",
    "email": "smita@gmail.com"
  },
  {
    "id": 2,
    "firstName": "Sia",
    "lastName": "Gupta",
    "email": "sia@gmail.com"
  },
  {
    "id": 3,
    "firstName": "Shivansh",
    "lastName": "Kumar",
    "email": "shivansh@gmail.com"
  },
  {
    "id": 4,
    "firstName": "Zara",
    "lastName": "Khan",
    "email": "zara@gmail.com"
  }
]
```

12) Run this on browser with the below given url

<http://localhost:3000/users>

13) Create new folder model and add new file user.model.ts

```
export class User{
  id: number;
  firstName: string;
  lastName: string;
  email: string;
}
```

14) Create new folder services inside app folder and then create user service by typing below command
ng g s services/user

and Add below code inside user.service.ts

```
import { Injectable } from '@angular/core';
import { HttpClient } from '@angular/common/http';
import { User } from '../model/user.model';

@Injectable()
export class UserService {

  constructor(private http:HttpClient) { }
  baseUrl:string = 'http://localhost:3000/users';

  // Get All Users
  getUsers(){
    return this.http.get<User[]>(this.baseUrl);
  }

  // Get User By Id
  getUserById(id: number){
    return this.http.get<User>(this.baseUrl+'/'+id);
  }

  // Create User
  createUser(user: User) {
    return this.http.post(this.baseUrl, user);
  }

  // Modify User
  updateUser(user: User) {
    return this.http.put(this.baseUrl + '/' + user.id, user);
  }
}
```

```
// Delete User
deleteUser(id: number) {
  return this.http.delete(this.baseUrl + '/' + id);
}
}
```

15) Create new folder images inside assets folder and add one image

16) Register your services and routing here, App.module.ts file

```
import { BrowserModule } from '@angular/platform-browser';
import { NgModule } from '@angular/core';

import { AppRoutingModule } from './app-routing.module';
import { AppComponent } from './app.component';
import { LoginComponent } from './login/login.component';
import { AddUserComponent } from './add-user/add-user.component';
import { EditUserComponent } from './edit-user/edit-user.component';
import { ListUserComponent } from './list-user/list-user.component';
import { ReactiveFormsModule } from '@angular/forms';
import { HttpClientModule } from '@angular/common/http';
import { UserService } from './services/user.service';
import { HomeComponent } from './home/home.component';

@NgModule({
  declarations: [
    AppComponent,
    LoginComponent,
    AddUserComponent,
    EditUserComponent,
    ListUserComponent,
    HomeComponent
  ],
  imports: [
    BrowserModule,
    AppRoutingModule,
    ReactiveFormsModule,
    FormsModule,
    HttpClientModule
  ],
  providers: [UserService],
  bootstrap: [AppComponent]
})
export class AppModule { }
```

17) Modify App.component.html with below code

18) Create [images](#) folder in [asest](#) folder and copy [/users.png](#) in it

[<!--The content below is only a placeholder and can be replaced.-->](#)

```
<div style="color:white; background-color: purple; padding: 10px; ">
  <h2 style="text-align:center; margin-top: 20px; ">User Management Angular App ...!
</h2>
</div>
<hr/>
```

```
<div class="container">
  <div class="row">
    <div class="col-md-8">
      <router-outlet> </router-outlet>
    </div>
    <div class="col-md-4">
      <br/>
      
    </div>
  </div>
  <br/> <br/>
  <div>
    <footer> <span class="text-danger"> <strong>This site is copyright &copy; to MY
DREAMZ COMPANY.
    All rights are reserved here ..!</strong> </span> </footer>
  </div>
</div>
```

19) Add below code inside [home.component.html](#) file

```
<br/> <br/>

<div class="col-md-6 col-md-offset-2">
  <h3>Angular </h3>
  <br/>

  <h5>In this demo, we will learn about </h5>
  <ul class="text-success" style="font-weight: bold">
    <li>
      Angular Architecture
      <ul>
        <li>Modules</li>
        <li>Components
          <ol>
            <li>Root Component or App Component</li>
          </ol>
        </li>
```

```
<li>Services and Dependency Injection</li>
<li>Models</li>
<li>Directives</li>
<li>Pipes, etc</li>
</ul>
</li>
</ul>

</div>
<a class="text-right" [routerLink]="['/login']"><h3>Login..</h3></a>
```

```
<br/><br/>
<div class="col-md-6 col-md-offset-2">
  <h3>Login</h3>
  <br/>
<form [formGroup]="loginForm" (ngSubmit)="onSubmit()">
  <div class="form-group">
    <label for="email">Email address:</label>
    <input type="email" class="form-control" formControlName="email" id="email">
    <div *ngIf="submitted && loginForm.controls.email.errors" class="error">
      <div class="text-danger" *ngIf="loginForm.controls.email.errors.required">Email is
required</div>
    </div>

  </div>
  <div class="form-group">
    <label for="pwd">Password:</label>
    <input type="password" class="form-control" formControlName="password"
id="pwd">
    <div *ngIf="submitted && loginForm.controls.password.errors" class="error">
      <div class="text-danger"
*ngIf="loginForm.controls.password.errors.required">Password is required</div>
    </div>
  </div>
  <button class="btn btn-success">Login</button> &nbsp; &nbsp;
  <button class="btn btn-danger" type="reset">Cancel</button>
<br/><br/>
  <div *ngIf="invalidLogin" class="error">
    <div class="text-danger">Invalid credentials.</div>
  </div>
</form>
</div>
```



```

import { Component, OnInit } from '@angular/core';
import { FormBuilder, FormGroup, Validators } from "@angular/forms";
import { Router } from "@angular/router";

@Component({
  selector: 'app-login',
  templateUrl: './login.component.html',
  styleUrls: ['./login.component.css']
})
export class LoginComponent implements OnInit {

  loginForm: FormGroup;
  submitted: boolean = false;
  invalidLogin: boolean = false;
  //constructor dependency injection
  constructor(private formBuilder: FormBuilder, private router: Router, ) { }

  onSubmit(){
    this.submitted = true;
    if(this.loginForm.invalid){
      return;
    }
    if(this.loginForm.controls.email.value == "smita@gmail.com" &&
      this.loginForm.controls.password.value == "password"){
      localStorage.setItem("username",this.loginForm.controls.email.value);
      this.router.navigate(['list-user']);
    }
    else{
      this.invalidLogin = true;
    }
  }

  ngOnInit() {

    this.loginForm = this.formBuilder.group({
      email: ['', Validators.required],
      password: ['', Validators.required]
    });

  }
}

```

22) Modify list-user.component.html

```

<br/><br/>
<div class="col-lg-12">
  <h2> User Details</h2>

```



```

constructor(private router: Router, private userService: UserService) { }

// Initialize with default list of users
ngOnInit() {
  if(localStorage.getItem("username")!=null){

    this.userService.getUsers()
      .subscribe(data=> {
        this.users = data;
      });
  }
  else
    this.router.navigate(['/login']);
}

// logOff user
logOutUser():void{
  if(localStorage.getItem("username")!=null){
    localStorage.removeItem("username");
    this.router.navigate(['/login']);
  }
}

// Delete User
deleteUser(user: User): void {
  let result = confirm('Do you want to delete the user?')
  if(result)
  {
    this.userService.deleteUser(user.id)
      .subscribe( data => {
        this.users = this.users.filter(u => u !== user);
      });
  }
};

// Modify User
editUser(user: User): void {
  localStorage.removeItem("editUserId");
  localStorage.setItem("editUserId", user.id.toString());
  this.router.navigate(['edit-user']);
};

// Add New User
addUser(): void {
  this.router.navigate(['add-user']);
};

```

```
}
```

24) Modify add-user.component.html

```
<br/><br/>
<div class="col-md-6 col-md-offset-2">
  <h2>Add User</h2>
  <br/>
  <form [formGroup]="addForm" (ngSubmit)="onSubmit()">
    <div class="form-group">
      <label for="firstName">First Name:</label>
      <input formControlName="firstName" placeholder="First Name" name="firstName"
class="form-control" id="firstName">
      <div *ngIf="submitted && addForm.controls.firstName.errors" class="error">
        <div class="text-danger"
*ngIf="addForm.controls.firstName.errors.required">FirstName is required</div>
      </div>
    </div>

    <div class="form-group">
      <label for="lastName">Last Name:</label>
      <input formControlName="lastName" placeholder="Last name" name="lastName"
class="form-control" id="lastName">
      <div *ngIf="submitted && addForm.controls.lastName.errors" class="error">
        <div class="text-danger"
*ngIf="addForm.controls.lastName.errors.required">LastName is required</div>
      </div>
    </div>

    <div class="form-group">
      <label for="email">Email address:</label>
      <input type="email" formControlName="email" placeholder="Email" name="email"
class="form-control" id="email">
      <div *ngIf="submitted && addForm.controls.email.errors" class="error">
        <div class="text-danger" *ngIf="addForm.controls.email.errors.required">Email is
required</div>
      </div>
    </div>

    <button class="btn btn-success">Save</button> &nbsp; &nbsp; &nbsp;
    <button class="btn btn-danger" type="reset">Cancel</button>
&nbsp; &nbsp; &nbsp; &nbsp; &nbsp;
    <a class="btn btn-warning" href="/list-user">Back to List</a>

  <br/><br/>
</form>
</div>
```

25) Modify add-user.component.ts

```
import { Component, OnInit } from '@angular/core';
import { FormGroup, FormBuilder, Validators } from '@angular/forms';
import { Router } from '@angular/router';
import { UserService } from '../services/user.service';

@Component({
  selector: 'app-add-user',
  templateUrl: './add-user.component.html',
  styleUrls: ['./add-user.component.css']
})
export class AddUserComponent implements OnInit {

  addForm: FormGroup;
  submitted: boolean = false;

  constructor(private formBuilder: FormBuilder, private router: Router,
    private userService: UserService) { }

  ngOnInit() {
    this.addForm = this.formBuilder.group({
      id:[],
      firstName: ['', Validators.required],
      lastName:['', Validators.required],
      email: ['', Validators.required]
    });
  }

  onSubmit() {
    this.submitted = true;
    if(this.addForm.invalid){
      return;
    }
    this.userService.createUser(this.addForm.value)
      .subscribe( data => {
        this.router.navigate(['list-user']);
      });
  }
}
```

26) Modify edit-user.component.html

```
</br><br>  
<div class="col-md-6 col-md-offset-2">  
  <h2>Edit User      &nbsp; &nbsp; &nbsp; &nbsp; &nbsp; &nbsp;
```

27) Modify edit-user.component.ts

```
import { Component, OnInit } from '@angular/core';
import { User } from '../model/user.model';
import { FormGroup, FormBuilder, Validators } from '@angular/forms';
import { Router } from '@angular/router';
```

```

import { UserService } from '../services/user.service';
import { first } from 'rxjs/operators';

@Component({
  selector: 'app-edit-user',
  templateUrl: './edit-user.component.html',
  styleUrls: ['./edit-user.component.css']
})
export class EditUserComponent implements OnInit {

  user: User;
  editForm: FormGroup;
  submitted: boolean = false;
  constructor(private formBuilder: FormBuilder, private router: Router,
    private userService: UserService) { }

  ngOnInit() {
    if(localStorage.getItem("username")!=null){
      let userId = localStorage.getItem("editUserId");
      if(!userId) {
        alert("Invalid action.")
        this.router.navigate(['list-user']);
        return;
      }
      this.editForm = this.formBuilder.group({
        id: [],
        email: ['', Validators.required],
        firstName: ['', Validators.required],
        lastName: ['', Validators.required]
      });

      this.userService.getUserById(+userId)
        .subscribe( data => {
          this.editForm.setValue(data);
        });
    }
    else
      this.router.navigate(['login']);
  }

  onSubmit() {
    this.submitted = true;
    if(this.editForm.invalid){
      return;
    }
    this.userService.updateUser(this.editForm.value)
      .pipe(first())
      .subscribe(

```

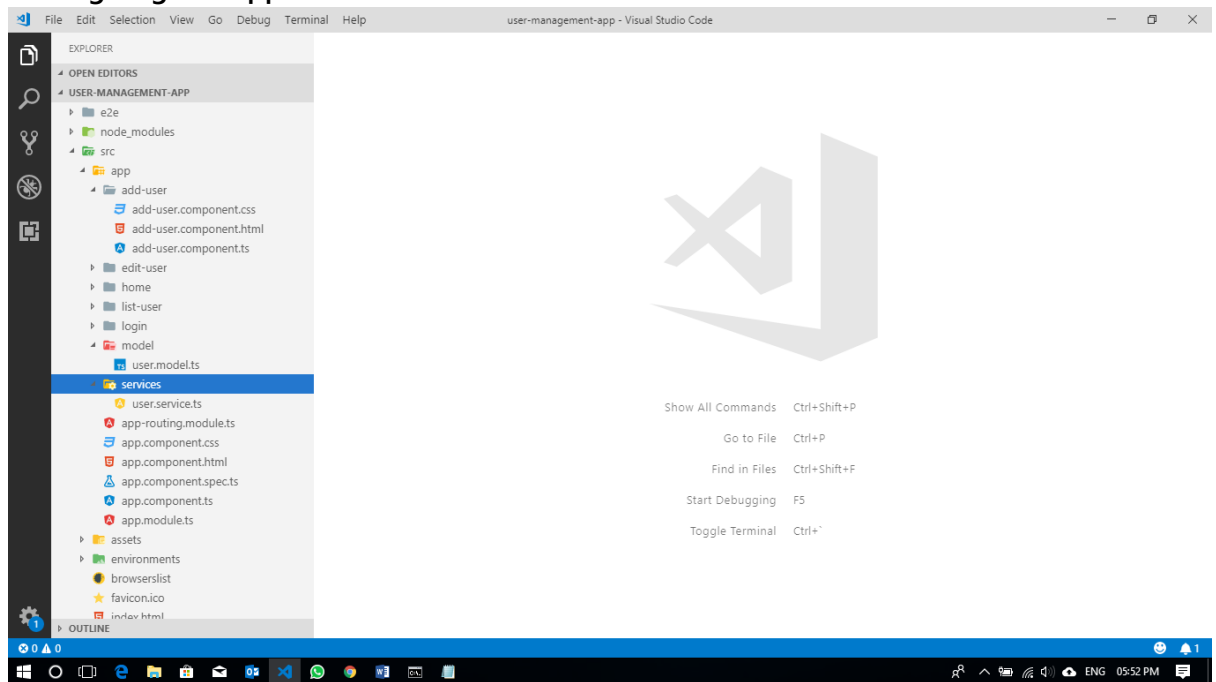
```

data => {
  this.router.navigate(['list-user']);
},
error => {
  alert(error);
});
}

// logOff user
logoutUser():void{
  if(localStorage.getItem("username")!=null){
    localStorage.removeItem("username");
    this.router.navigate(['/login']);
  }
}
}
}

```

28) Testing Angular Application



json-server --watch db.json

ng serve

User Management Angular App ...!

Angular

In this demo, we will learn about

- Angular Architecture
 - Modules
 - Components
 1. Root Component or App Component
 - Services and Dependency Injection
 - Models
 - Directives
 - Pipes, etc

Login..



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User Management Angular App ...!

Login

Email address:

Password:

Login

Cancel



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Password:



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User Management Angular App ...!

list-user works!

User Details

First Name	Last Name	Email	Action
SMITA	kumar	Smita@gmail.com	Delete Edit
SIA	gupta	Admin@gmail.com	Delete Edit
SHIVANSH	kumar	User@gmail.com	Delete Edit
ZARA	khan	Zara@gmail.com	Delete Edit



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UserManagementApp

localhost:4200/add-user

User Management Angular App ...!

Add User

First Name:

First Name

Last Name:

Last name

Email address:

Email

Save

Cancel

Back to List



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Windows Taskbar

System Tray

User Management Angular App ...!

list-user works!

Add User

LogOff

User Details

First Name	Last Name	Email	Action
SMITA	kumar	Smita@gmail.com	Delete Edit
SIA	gupta	Admin@gmail.com	Delete Edit
SHIVANSH	kumar	User@gmail.com	Delete Edit
ZARA	khan	Zara@gmail.com	Delete Edit



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User Management Angular App ...!

Edit User

Log Off

First Name:

smita

Last Name:

kumar

Email address:

smita@gmail.com

Update

Cancel

